Revised Proposal: Mechanistic Carbon-Cycle Box Model

02-120 final project · @kar · Nov 8, 2026

Questions

Question 1. Can a carbon-cycle model built from physical and chemical rules, with no curve fitting to atmospheric CO_2, reproduce the rise measured at Mauna Loa and across NOAA's global network from historical emissions?

Question 2. I will build an interactive box model with four reservoirs: the atmosphere, the surface ocean, the deep ocean and the land biosphere. Carbon moves between them by stated mechanisms: air–sea gas exchange driven by ocean carbonate chemistry, slow mixing into the deep ocean, and plant uptake that grows with CO_2. Users can change emissions, the time period and the physical parameters, such as ocean mixing speed or plant response, and watch how the atmosphere, ocean and land respond.

Question 3. I will check the model against published results it was not tuned to. These are the 100 GtC pulse result of Joos et al. (2013), the Global Carbon Budget's measured ocean and land sinks, and the observed ocean buffer (Revelle) factor. Only after these checks will I compare the model with the CO_2 record.

What changes from the first proposal

The first plan used the Joos et al. (2013) impulse response function: a sum of four exponentials, $IRF(t) = a_0 + \Sigma\, a_i e^{(-t/\tau_i)}$. Its seven numbers are a curve fit to the output of other models. Nothing in the formula says why carbon leaves the air, so it can't explain its own errors.

Running it showed the problem. Driven by all emissions since 1850, it predicts 478 ppm for 2024, while Mauna Loa measured 424.6 ppm. The formula can't say whether the ocean, the land or the emissions data is responsible.

The revised model replaces the fitted curve with mechanisms:

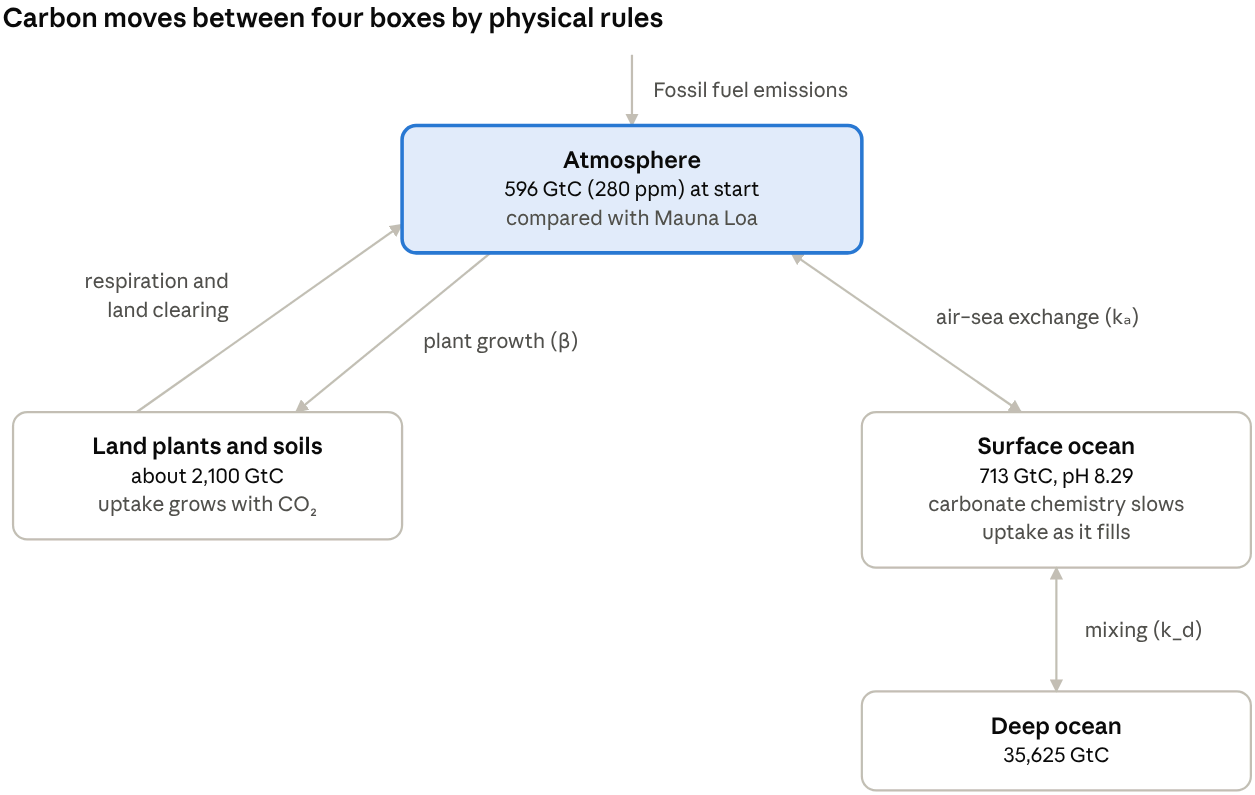
- **Ocean uptake** comes from the chemistry of dissolved carbon, so the ocean takes up less as it fills. A fixed exponential can't do this.
- **Land uptake** comes from plant growth that rises with CO_2 and from soil respiration.

- **Every parameter is a measured physical quantity**, such as ocean volume, gas transfer speed or chemical equilibrium constants. None is tuned to the CO_2 record.

The Joos result becomes a test the model must pass, not the model itself.

Scientific model

The model is a four-box carbon cycle. The three ocean and atmosphere boxes follow BEAM (Bolin and Eriksson Adjusted Model) of Glotter et al. (2014). I add a land box with CO_2 fertilization. All four are solved as ordinary differential equations in GtC.



four-box carbon cycle · starting stocks from Glotter et al. (2014)

Emissions enter the atmosphere. The ocean and land take carbon back at rates set by chemistry, mixing and plant growth.

Atmosphere, surface ocean, deep ocean (BEAM):

```
\frac{dM_{AT}}{dt} = E_{fossil}(t) + E_{LUC}(t) - k_a\,(M_{AT} -
A\,B\,M_{UP}) - F_{land}
```

$$\frac{dM_{UP}}{dt} = k_a\,(M_{AT} - A\,B\,M_{UP}) - k_d\,\left(M_{UP} - \frac{M_{LO}}{\delta}\right)$$

$$\frac{dM_{LO}}{dt} = k_d\,\left(M_{UP} - \frac{M_{LO}}{\delta}\right)$$

The key mechanism is the factor B, the share of surface-ocean carbon present as dissolved CO_2. It comes from ocean chemistry: $B = 1 / (1 + k_1/[H^+] + k_1k_2/[H^+]^2)$. The hydrogen-ion concentration $[H^+]$ is re-solved every time step with alkalinity held constant. As the ocean takes up carbon it becomes more acidic, B rises, and the ocean pushes back. That is why uptake slows over time without any fitted constant.

Land biosphere (added):

$$F_{land} = NPP_0\left(1 + \beta \ln\frac{M_{AT}}{M_{AT,0}}\right) - \frac{M_L}{\tau_L}, \qquad \frac{dM_L}{dt} = F_{land} - E_{LUC}(t)$$

Plants grow faster as CO_2 rises, through the logarithmic β term. Soils and dead plants return carbon at a rate proportional to the land carbon stock. Land-use emissions are taken out of the land box, so clearing forests both emits CO_2 and shrinks the future sink.

Parameter	Meaning	Value	Source
k_a	Air–sea exchange rate	0.2 yr^{-1} (5-yr timescale)	Bolin & Eriksson (1959), via Glotter et al. (2014)
k_d	Surface–deep ocean mixing rate	0.05 yr^{-1} (20-yr timescale)	Bolin & Eriksson (1959), via Glotter et al. (2014)
δ	Deep-to-surface ocean volume ratio	50	Bolin & Eriksson (1959), via Glotter et al. (2014)
k_1, k_2	Carbonic acid dissociation constants	8.00×10^{-7}, 4.53×10^{-10} mol/kg	Mehrbach et al. (1973), via Glotter et al. (2014)
k_H	CO_2 solubility (Henry's law)	1.23×10^{3}	Weiss (1974), via Glotter et al. (2014)
Initial state	Atmosphere / surface / deep ocean, pH	596 / 713 / 35,625 GtC, pH 8.29	Glotter et al. (2014), Table 4
NPP_0	Pre-industrial global plant growth	~60 GtC/yr	Enting et al. (1994), Model J

Parameter	Meaning	Value	Source
β	CO_2 fertilization strength	0.38–0.4 (range to explore)	Enting et al. (1994)
τ_L	Land carbon turnover time	~35 yr (approximate, to confirm)	Land stock ÷ NPP_0

The model will be solved with a 0.01-year time step. Glotter et al. report that a 0.1-year step makes pH oscillate when emissions change quickly.

Checks against published results

The main check reproduces the Joos et al. (2013) pulse experiment with the new model: 25 ± 9% of a 100 GtC pulse should remain after 1,000 years. The model was not built from that result, so passing it means something. A quick prototype already gives **25%**.

Three more checks test each reservoir separately. Matching the total CO_2 alone isn't enough, since errors in the ocean and land can cancel.

Check	Published value	Source	Prototype (β = 0.4)
100 GtC pulse still airborne at 1,000 yr	25 ± 9%	Joos et al. (2013)	25% (passes)
Ocean sink, 2015–2024 average	3.2 ± 0.4 GtC/yr	Global Carbon Budget 2025	1.0 GtC/yr (fails)
Land sink, 2015–2024 average	2.4 ± 0.8 GtC/yr	Global Carbon Budget 2025	5.3 GtC/yr (fails)
Anthropogenic carbon in the ocean by 1994	118 ± 19 GtC	Sabine et al. (2004)	36 GtC since 1850 (fails)
CO_2 in 2024	424.6 ppm (Mauna Loa)	NOAA GML	418 ppm (close)

The prototype gets the 2024 total within 7 ppm, but for the wrong reason. Its ocean takes up about a third of the measured amount, and its land makes up the difference. The first proposal could not have shown this.

The project will trace the gap to a mechanism. Candidates: BEAM's 1959-era mixing rates (k_a, k_d), its small surface-ocean box, and the strength of plant fertilization. Each is a physical quantity with independent estimates, so changes will be justified by measurements, not by fitting the CO_2 curve.

The prototype numbers come from a first draft of the code and will be re-verified before they are reported.

Data

All inputs and checks use public data. Emissions drive the model; everything else is held back for checking.

Dataset	What it gives	Role
Global Carbon Budget 2025, national emissions	Fossil CO_2 for each country, plus global land-use emissions, yearly to 2024	Input; lets users switch off or scale single countries
Global Carbon Budget 2025, global budget table	Measured ocean sink, land sink and atmospheric growth for each year since 1959	Year-by-year check of each box
Scripps Mauna Loa record	Monthly CO_2 since 1958	Check of the atmosphere
NOAA GML global marine surface mean	Global average CO_2 since 1979 from remote sea-level sites worldwide (Mauna Loa excluded)	Independent global check
Law Dome ice core (MacFarling Meure et al., 2006)	CO_2 in air trapped in Antarctic ice, covering 1850–1958 and earlier	Check for the century before direct measurements
Sabine et al. (2004)	Ocean inventory of human-emitted carbon in 1994, from global ship surveys	Check of the ocean boxes

None of these is used to set a parameter. The model's numbers come only from the physical sources in the parameter table.

Interactive tool

The web page from the first version is kept, with its simulation engine replaced by the box model. Users will be able to:

- **Watch the carbon move.** A live diagram of the four boxes shows each stock and flow, year by year.
- **Change the physics.** Sliders for ocean mixing speed, air–sea exchange and plant fertilization show which mechanism controls the atmosphere.
- **Change the emissions.** Scale all emissions, turn off land-use change, remove a country, or extend to 2100 under a chosen scenario.
- **See every check live.** The pulse test, both sinks and the CO_2 record update with each change, marked pass or fail.
- **Import their own data**, as the current page already allows.

The notebook will hold the same model in Python, with tests that compare its output against the checks above.

Plan

1. Implement the four-box model in Python and confirm the steady state holds with zero emissions.
2. Run the pulse test and the sink checks; report pass or fail for each.
3. Investigate the ocean shortfall, using independent estimates of ocean mixing rates.
4. Run the historical simulation against Mauna Loa, the NOAA global mean and Law Dome.
5. Port the model to the interactive page and add the box diagram and live checks.
6. Write up the results, including which mechanisms matter most.

Sources

- Glotter, M., Pierrehumbert, R., Elliott, J., Matteson, N., & Moyer, E. (2014). A simple carbon cycle representation for economic and policy analyses. *Climatic Change*, 126, 319–335.
- Joos, F., et al. (2013). Carbon dioxide and climate impulse response functions for the computation of greenhouse gas metrics: a multi-model analysis. *Atmos. Chem. Phys.*, 13, 2793–2825.

- Friedlingstein, P., et al. (2026). Global Carbon Budget 2025. *Earth Syst. Sci. Data*, 18, 3211–3288.
- Sabine, C. L., et al. (2004). The oceanic sink for anthropogenic CO_2. *Science*, 305, 367–371.
- Enting, I. G., Wigley, T. M. L., & Heimann, M. (1994). Future emissions and concentrations of carbon dioxide. CSIRO Division of Atmospheric Research Technical Paper 31.
- MacFarling Meure, C., et al. (2006). Law Dome CO_2, CH_4 and N_2O ice core records. *Geophys. Res. Lett.*, 33, L14810.
- NOAA Global Monitoring Laboratory. Calculation of global means.